A Multibranch Embedding Network With Bi-Classifier for Few-Shot Ship Classification of SAR Images

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***Abstract*— Ship classification in synthetic aperture radar (SAR) images is a challenge in the field of ocean monitoring. On the one hand, there are few labeled samples in SAR remote sensing ship datasets, and a commonly used single classification criterion cannot effectively represent the distribution of categories. On the other hand, the small size of the SAR ship and the inconspicuous appearance characteristics lead to the fact that the SAR ship samples are with less discriminative information; therefore, the rich feature space of a ship cannot be effectively obtained, which increases the difficulty of target distinguishability. A multibranch embedding network with bi-classifier (MBEN-BC) model was proposed to address these problems and for few-shot SAR ship classification. First, the MBEN module was utilized to extract the multiscale feature map spatial information of the input image at multiple levels and establish cross-channel information interaction so as to obtain discriminative features at the local and global levels, which effectively enriched the feature space. Then, the BC module was constructed to represent the image features from the image level and descriptor level, respectively, and the two classification criteria were presented to promote a more compact distribution of similar samples in the feature space in order to effectively represent the distribution of categories with a small number of labeled samples. Experimental validation was carried out using the FUSAR-Ship, Moving and Stationary Target Acquisition and Recognition (MSTAR) dataset, and OPENSAR-Ship dataset, and the MBEN-BC method achieved superior performance and good generalization ability compared to the current popular and state-of-the-art few-shot methods.**

***Index Terms*— Bi-classifier (BC), metric learning, multibranch embedding network (MBEN), synthetic aperture radar (SAR) ship classification.**

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I. Introduction

COMPARED with optical sensors, synthetic aperture radar (SAR) imaging is not limited by time and weather conditions and can provide high-resolution day-night images and without the influence of weather, which is one of the most effective means for maritime target surveillance [1]. Among them, ships are important carriers of maritime transportation, and their identification and classification have become an important task at present. Whether in the military aspects, such as precision strikes, surveillance, and antiterrorism, or the civilian aspects, such as sea traffic control, antismuggling, and search and rescue, SAR ship image classification has extensive applications [2].

In recent years, since the rapid progress of deep learning (DL) technology, various network models have emerged, many of which have been used in the field of SAR object classification [3]. For instance, convolutional neural network (CNN) [4], [5], [6], [7], [8], [9] can effectively capture the local feature information of texture, edges, etc. in SAR images by virtue of parameter sharing, translation invariance, and other unique convolutional processing. Together with the advantages of simplicity, intuition, and extensibility, CNN is popular in the field of SAR image classification. Transformer-based network models, especially variational inference with adversarial learning for end-to-end text-to-speech and Swins [10], [11], [12], use a multilayer self-attention mechanism to globally model the entire input sequence to capture global information in SAR images and are widely sought after by researchers for their advantages of long-range dependent modeling and parallelizable computation. In addition, significant progress has been made in graphical neural networks (GNNs) [13], [14]. GNNs can efficiently build graph structures for SAR targets of different shapes and sizes and are also capable of fusing multiscale feature information through multilevel graph convolution operations to enhance the model's classification precision for targets.

DL methods, however, often require the construction of large-scale neural networks and need to depend on massive amounts of labeled data to train the models for the sake of avoiding the overfitting problem; however, for some special types of SAR targets, for instance, ships, the image numbers are relatively small, which makes it hard to acquire abundant labeled data. The few-shot learning (FSL) technique, therefore,

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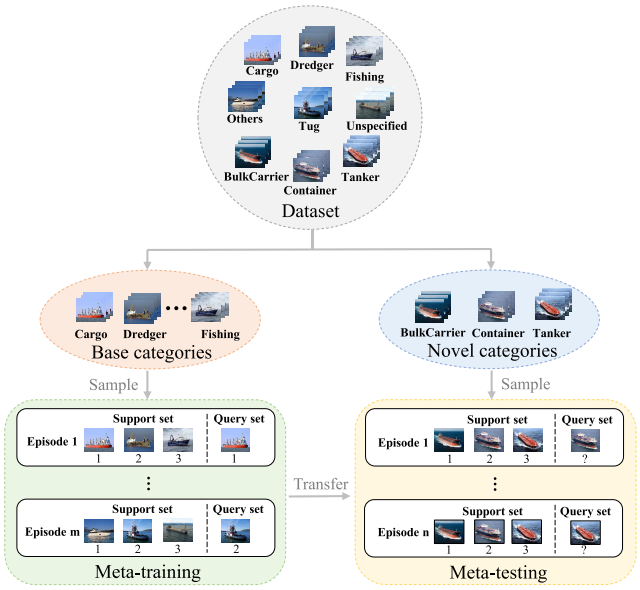


Fig. 1. Training process of meta-learning. The base categories and novel categories do not intersect. Each eposide has a C-way K-shot task (the above represents a three-way one-shot task).

becomes a new direction to solve this problem, and more and more researchers have begun to turn their attention to few-shot classification and study how to achieve the same classification effect as the massive samples under limited samples. At present, A large number of studies have focused on SAR image classification, which involves the FSL methods mainly include data-augmentation-based [15], [16], [17], [18], [19], [20], transfer-learning-based [21], [22], [23], [24], [25], [26], and meta-learning-based [27], [28], [29], [30], [31], [32], [33], [34], [35], [36], [37], [38], [39], [40], [41], [42], [43], [44], [45], [46]. One should note that FSL for ship classification is relatively rarely studied. Several known investigations include Zhang et al. [53], who incorporated scattering characteristics into network learning based on the scattering point layout (SPT) of key scattering points, constructing the scattering point topology for few-shot ship classification (SPT-FSC). Li et al. [54] proposed a cross-domain FSL method for fine-grained SAR ship recognition (CDFS-FG), using fine-grained features to obtain rich contextual information. Gao et al. [55] proposed a new method of FSL for SAR image recognition consisting of multifeature fusion network (MFFN) and weighted distance classifier (WDC) to solve the problem of large internal class diversity and high cross-class similarity in SAR images. Yang et al. [45] proposed a hybrid loss graph attention network [mixed loss graph attention network (MGA-Net)] to address the problem of insufficient training samples for automatic target classification. In summary, from the studies that have been done, it can be found that the meta-learning method is the most popular FSL because it requires fewer labeled samples than other methods and can show good performance on small-scale SAR image datasets.

In principle, the main idea of meta-learning is to let the model to learn generalized learning strategies or optimization algorithms from limited training samples so that it can adapt quickly and make correct predictions when facing a new task. The process of meta-learning includes meta-training and meta-testing, and the Episode training mechanism [27] is often used for end-to-end training so that it simulates the execution process of real tasks, making the training closer to the actual application scenarios. As exhibited in Fig. 1, the C-way K-shot task is demonstrated in the meta-training phase, i.e., each episode randomly samples C categories in the base categories, each category collects K-labeled samples to compose the support set, and a small portion of the remaining samples in every category compose the query set, and then the model learns to extract meta-knowledge between different tasks through iterative training of multiple episodes. The same training mechanism as in the training phase is used in the meta-testing phase, and directly applying the trained model to novel categories of classification tasks.

The meta-learning can be grouped into two main subcategories: optimization-based and metric-based. The former methods are divided into two main ways: one is to optimize the initial representation of the model, and the other is to learn an efficient optimizer. For example, Finn et al. [28] used a Model-Agnostic Meta-Learning network to capture the initialization state of subtle parameter changes and swiftly adjust to a new task through the minimum gradient descent algorithm. Ravi and Larochelle [29] used the long short-term memory network to autonomously initialize and update parameters when facing new tasks. Bertinetto et al. [30] adopted ridge regression and logistic regression as parameter optimization methods for classifiers and were able to quickly find optimal solutions with high generalization ability. Overall, although optimization-based methods can learn on a relatively small number of samples and adapt quickly to new tasks, they tend to rely on optimization algorithms, as well as be sensitive to initial conditions, which may arouse the learning process to entrap a locally optimal solution if the optimization algorithms and initial conditions are not properly chosen, affecting the final classification performance.

The metric-based approach mainly consists of two processes, i.e., feature extraction and category reasoning. To be specific, the features are first learned and extracted using a feature extractor, and then the distances among the query features and the support features are computed and category inference is performed using various metrics, such as the cosine similarity metric [27], Euclidean distance metric [31], neural network-based learnable metric [32], covariance metric [33], divergence distribution metric [34], Earth Mover's distance metric [35], Brownian distance covariance metric [36], and so on. Based on the foundational idea, some known methods combining both of these processes to achieve better inference performance have been further developed in recent years. For instance, Dong et al. [37] introduced situational attention modules to adaptively select important local regions in tasks. Kang et al. [38] used autocorrelation representation and interrelated attention mechanisms to collaboratively learn relevant embeddings that emphasize the target region for better classification. Gao et al. [39] used a lightweight adaptive task attention module to boost the adaptive capture ability of target features and used a bi-prototype module to build high-precision prototypes to improve classification performance. In conclusion, the metric-based approach is very efficient and convenient. This approach avoids refine-tuning

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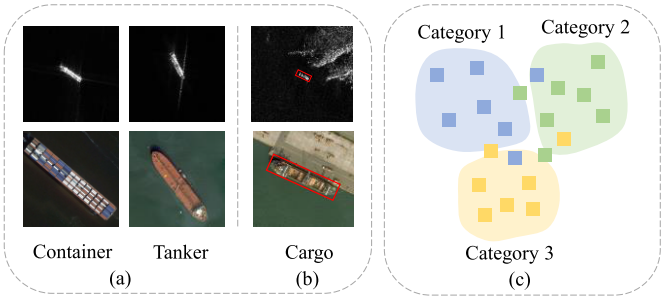


Fig. 2. Two issues of SAR ship classification. The first box represents the difference among SAR ships and optical ships. (a) High visual similarity. (b) Small target size. The second box (i.e. (c)) indicates inaccurate distribution of SAR target categories.

in new tasks and uses the model directly by keeping the original parameters unchanged. Compared with the above approaches, the metric-based method has become a more promising approach for FSL.

Although current metric-learning methods are encouraging, most of them were initially proposed for target classification task of optical images. In terms of SAR ship classification, however, they still suffer from the following several problems.

1) Feature Extraction: As shown in Fig. 2(a) and (b), SAR ship images are grayscale maps as usual, the differences in the external features of different ships are not as clear as in optical images, and there is a high visual similarity between image samples. Ship targets are, moreover, usually smaller in SAR images, resulting in the inclusion of less target discriminability information. Existing methods, however, usually only use each layer of the feature extractor to extract a single-scale image feature map, which cannot obtain a rich feature space for SAR ship samples with less target information, increasing the difficulty of target distinguishability.

One effective way to figure out this issue is to use the multiscale features of the ship target. In fact, some progress has been made in certain other fields of image classification. For instance, Chen et al. [40] constructed a multiscale adaptive task attention network to generate multiscale features by adopting four average pooling layers, and then choosing specific tasks at multiple scales to achieve optical image classification. Wang et al. [41] considered extracting the complementary features of convolutional layers, and a multiscale prototype network was proposed to enhance the discrimination of feature representation. Although they were not used to address SAR ship classification, they suggest that the idea of extracting multiscale features of targets can effectively figure out the problem of insufficient discriminative information of SAR ship targets.

2) Metric Classification: As shown in Fig. 2(c), the difficulty of sampling the SAR ship dataset results in relatively limited training samples and categories. In addition, existing methods usually rely only on a single classification criterion to recognize images, exacerbating the problem that the target category inference is not robust enough to effectively represent the category distribution of insufficient labeled samples, thus increasing the difficulty of few-shot SAR ship images classification.

From this, it can be inferred that if samples can be well classified through different metric modules simultaneously, similar samples can be more compactly distributed in the feature space, thus enabling more effective and accurate identification of target categories. Li et al. [42] constructed a bi-similarity network to combine different metric modules to enhance the classification capacity of optical images. Zeng et al. [43] used two relation modules to obtain the similarity scores of first- and second-order features between hyperspectral images, thereby improving classification performance. These examples validate that using multiple metric modules simultaneously for target classification is a feasible and effective strategy.

According to the above analysis, a multibranch embedding network with the bi-classifier model (MBEN-BC) is proposed for few-shot SAR ship image classification. In terms of feature extraction, so as to effectively extract rich and discriminative features, the MBEN module uses each layer of the feature extractor to extract image feature maps at multiple scales in a multibranching manner as a means of obtaining multiscale spatial information. Then, using cross-channel attention weights to establish channel dependencies, more information interaction is obtained between local and global attention, thus effectively enriching the feature space. In terms of metric classification, for the sake of recognizing the target category more effectively and accurately, bi-classifiers (BC) [i.e., image-level classification (IC) module and descriptor-level classification (DC) module] are used to perform image feature representation from the image level and descriptor level, respectively. Then, the first- and second-order features of the support and query samples are extracted. Next, the relationship scores of the two features are computed and fused as a way to promote a more compact distribution of similar samples in the feature space, which further enhances the classification performance of SAR ship images.

Overall, the main contributions are summarized as follows.

1) A new MBEN-BC framework is proposed to realize effectively few-shot SAR ship image classification.
2) The MBEN module and the BC module are constructed. The former can extract the multiscale feature map space information to acquire discriminative features at both local and global levels. The latter performs image feature representations at the image level and descriptor level to promote a more compact distribution of similar samples.
3) Experimental validation using the FUSAR-Ship and Moving and Stationary Target Acquisition and Recognition (MSTAR) dataset shows that the MBEN-BC method achieves superior performance and good generalization ability compared to the current popular and advanced FSL methods.

The remainder of this study is organized as follows. The MBEN-BC method in detail is presented in Section II. Section III carries out the comparative, ablation and visualization experiments, respectively, and analyzes and describes them accordingly. Section IV summarizes the conclusions of the article.

II. Proposed Methodology

A. Overall Architecture of the MBEN-BC

The MBEN-BC framework is exhibited in Fig. 3, which is composed of the beneath modules: the MBEN module is the feature extractor, and the BC makes up of the IC module

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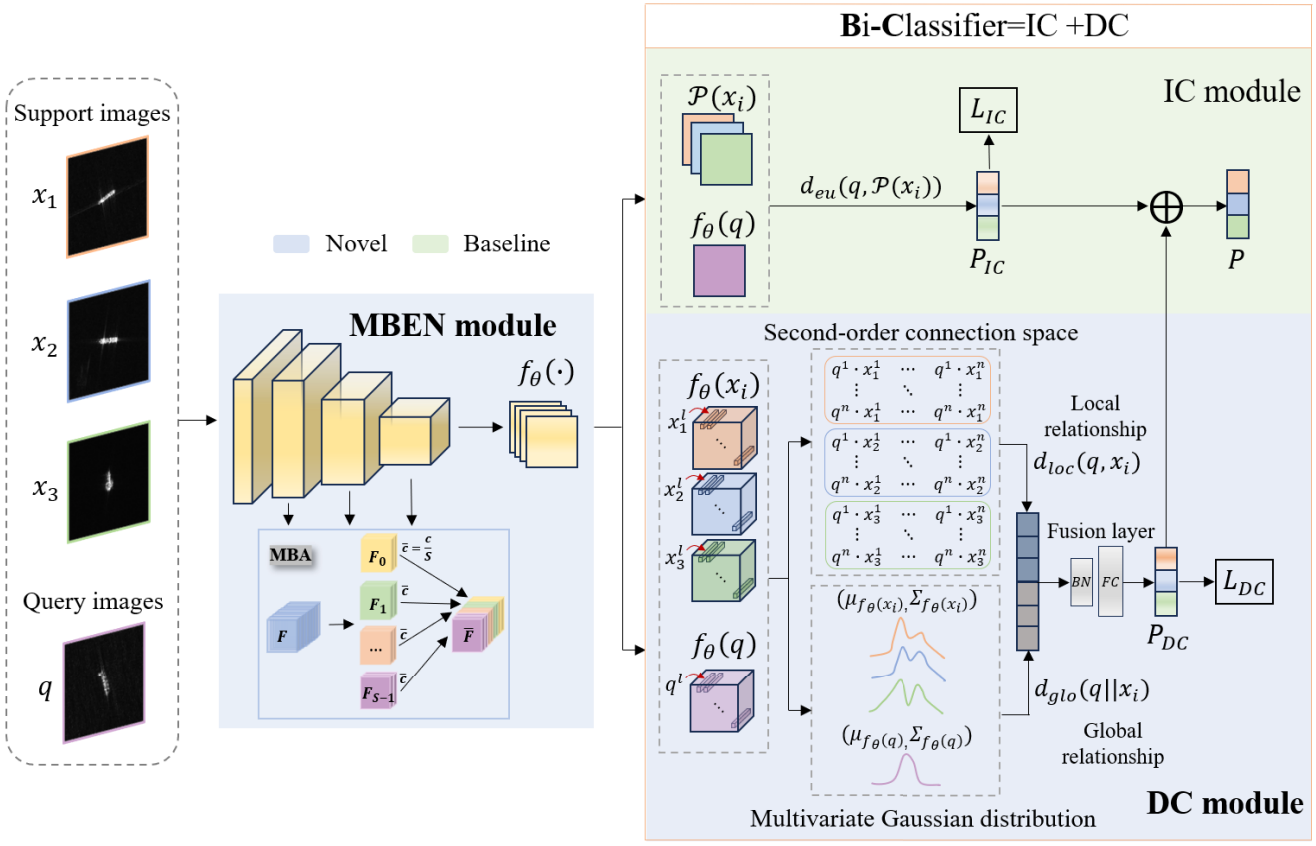


Fig. 3. Overall framework of the MBEN-BC. $f_\theta(\cdot)$ is feature maps with distinguishing information, $\mathcal{P}$ is support class prototype. d_{EU} is Euclidean distance metric, P_{IC}, P_{DC} are prediction scores, L_{IC}, L_{DC} are loss functions, d_{loc} is the local-relationship, d_{glo} is the global-relationship.

and the DC module. The flow of MBEN-BC model is chiefly shown below.

First, the support image and the query image are sent to the MBEN network simultaneously. The differentiation of the feature representation is enhanced by fusing calibrated multiscale features from different layers to extract more feature maps $f_\theta(\cdot)$ with discriminative information.

Next, $f_\theta(\cdot)$ is input to the IC module and DC module, respectively. In the IC module, the Euclidean distance metric d_{EU} is used to figure up the distance between the queries and the support class prototype $\mathcal{P}$ to obtain the prediction score P_{IC}.

Then, in the DC module, the feature map is first represented as a depth descriptor. The second-order connection space is established for the support class, and the queries and the image-to-class metric is used to capture the local relationship d_{loc} between the two; moreover, the multivariate Gaussian distribution is established for the support class, and the queries and the divergence distribution metric is used to acquire the global relationship d_{glo} between the two. Then, the prediction score P_{DC} is obtained through a fusion layer.

Finally, the two prediction scores are summed and fused to gain the final classification result; moreover, to illustrate the whole process of MBEN-BC more clearly, Algorithm 1 is shown above.

B. MBEN Module

Feature extraction is the key step in realizing SAR image classification. CNNs are widely used as feature embedding models at this stage due to their end-to-end autonomous learning properties; however, CNN treats all regions equally in an image, both at the channel level and at the spatial level, rather than focusing on representative regions. The inherent defect of CNN leads to its limited learning capability. With the continuous development and application of the attention mechanism, more and more image classification methods have begun to use the attention mechanism to obtain key features. Most of the existing methods can, however, only effectively capture the local information of single-scale image features and cannot effectively establish remote channel dependencies or usually stack multiple networks, bringing higher model complexity, and thus failing to adapt to few-shot image classification. In view of this, we propose the MBEN network for extracting multiscale feature map spatial information at multiple levels of an image and establishing cross-channel information interaction, which can effectively capture discriminative features at both local and global levels.

The training sample numbers available for the SAR dataset are relatively small compared to the large-scale optical image dataset. In order to decrease the risk of overfitting we, therefore, chose the Conv4 network as the backbone. As shown in Fig. 4, the main components of each layer of the Conv4 network and the MBEN network are demonstrated. The MBEN network mainly replaces the convolutional layers from the second to the fourth of Conv4 with multibranch attention (MBA) modules to achieve a good balance between feature extraction, model complexity and performance. The output of the previous layer is used as the input of the subsequent layer,

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Algorithm 1 MBEN-BC Training

Input: $\mathbb{T}$: tasks assignment, ξ: learning rate, $\lambda_1(\lambda_2)$: normalized weight, α: trade-off hyper-parameter

Output: network parameter η;

initialize randomly η;
repeat
  sample batch of tasks $t \sim \mathbb{T}$;
  for each task $t_i \in t$ **do**
    sample $\{D^s = \{(x_i, y_i)\}_{i=1}^{CK}$, $D^q = \{(q_i, y_i^q)\}_{i=1}^{CQ}\} \sim t_i$;
    MBEN: acquire embeddings of D^s and $D^q \rightarrow$ feature maps: $F(x_i), F(q_i) \rightarrow$ use three MBA modules $\rightarrow$ representative feature maps: $f_\theta(x_i), f_\theta(q_i)$;
    IC: $f_\theta(x_i) \xrightarrow{avg}$ support prototypes: $\mathcal{P}(x_i)$, query predictions: $P_{IC} = d_{eu}(f_\theta(q_i), \mathcal{P}(x_i)) \rightarrow L_{IC}$;
    DC: $f_\theta(x_i), f_\theta(q_i) \rightarrow$ depth descriptors: $[x_i^1, \ldots, x_i^n], [q_i^1, \ldots, q_i^n] \rightarrow$ local/global relationship: $d_{loc}(q_i, x_i), d_{glo}(q_i || x_i) \rightarrow$ query predictions: $P_{DC} = FC(BN(Cat(d_{glo}, d_{glo}))) \rightarrow L_{DC}$;
    Query predictions: $P = \lambda_1 * P_{IC} + \lambda_2 * P_{DC}$;
    $L = L_{IC} + \alpha L_{DC}$;
    Update network: $\eta \leftarrow \hat{u}\eta - \xi * \nabla_\eta L$;
  end for;
until convergence;

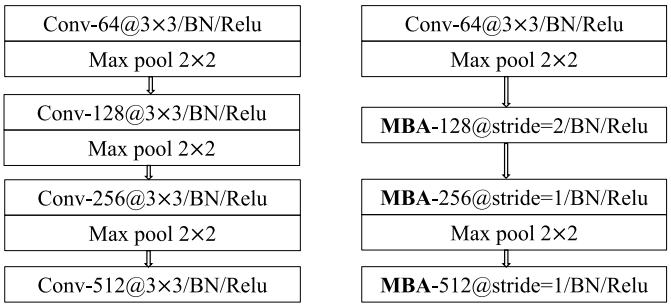


Fig. 4. Structural diagram of the Conv4 and the MBEN. (a) Conv4. (b) MBEN.

embedded layer-by-layer augmented feature representation. Each MBA module captures multiple different scales of information simultaneously through parallel operation of multiscale convolutional kernels, extracts the spatial information of different scale features of the image in a multibranch manner to amend the feature space, respectively. The channel attention weights are then used to construct cross-channel interactions. Next, Softmax operation is employed to recalibrate the corresponding channel attention weights to establish remote channel dependencies so that local-level and global-level attention interaction information can be better obtained. Finally, the differentiation of the feature representation is enhanced by fusing calibrated multiscale features from different layers, thus extracting more feature maps with discriminative information.

The framework of the MBA module is exhibited in Fig. 5. Specifically, $F \in R^{c\times h\times w}$ denotes the input feature map, where c, h, and w represent its input channel numbers height, and width. We extract the spatial location information at different scales of the input feature map by using a multibranching pattern, and every branch has an input channel dimension of c. For each branch, we compress the channel dimensions and generate different feature maps with spatial resolutions by using a multiscale convolutional kernel, thus efficiently extracting the spatial location information at diverse scales on every channel. As the multiscale convolutional kernel sizes grow, the parameter numbers, however, increase accordingly. In order to effectively preserve the details of the SAR image without increasing the computational cost, we introduce the grouped convolution method [47], where $\mathcal{G} = 2^{(k-1)/2}$, $\mathcal{G}$ denotes the group size and k denotes convolution kernel size; therefore, the generating function of the feature map F_j with different scales for different branches can be represented as

$$F_j = \mathrm{Conv}\big(k_j \times k_j, \mathcal{G}_j\big)(F) \tag{1}$$

where $F_j \in R^{\bar{c}\times h\times w}$, $j = 0, 1, \ldots, S-1$, S denotes the branch numbers, the output channel dimension of each branch: $\bar{c} = c/S$, and the jth group size $\mathcal{G}_j = 2^{(k_j-1)/2}$. The convolution kernel size varies across different feature maps to enhance the network's adaptability to diverse targets and fine details. This variation improves the model's generalization ability and resilience to noise, making it more robust across a range of tasks and conditions. Thus, the jth convolutional kernel size $k_j = 2j + 3$. The entire preprocessed multiscale feature map $\bar{F}$ is below

$$\bar{F} = \mathrm{Cat}\big(\big[F_0, F_1, \ldots, F_{S-1}\big]\big). \tag{2}$$

Next, the channel attention vectors $V_j \in R^{\bar{c}\times 1\times 1}$ at different scales are acquired through picking up the channel weight information of $\bar{F}$. In addition, for the sake of linking the context information of different scales and realizing the attention weight information interaction, it is necessary to fuse and splice the V_j of different scales; therefore, the entire multiscale channel attention vector V is gained in tandem form. The formula can be expressed as

$$\begin{aligned} V_j &= \sigma\big(w_1\delta\big(w_0\big(\mathrm{GAP}\big(F_j\big)\big)\big)\big) \\ V &= V_0 \oplus V_1 \oplus \cdots \oplus V_{S-1} \end{aligned} \tag{3}$$

where GAP represents global average pooling, δ denotes the linear correction function (ReLU), $w_0 \in \mathbb{R}^{\bar{c}\times(\bar{c}/r)}$ and $w_1 \in \mathbb{R}^{(\bar{c}/r)\times\bar{c}}$ denote the fully connected layers, r denotes the shrinkage rate, σ denotes the Sigmoid function, and $\oplus$ is the concatenation operator. The linear information between the channels can be linked more availably by using w_0 and w_1, which helps the information interaction among the high and low channels. Then, weights are assigned for channels by using σ, thus acquiring the information more accurately.

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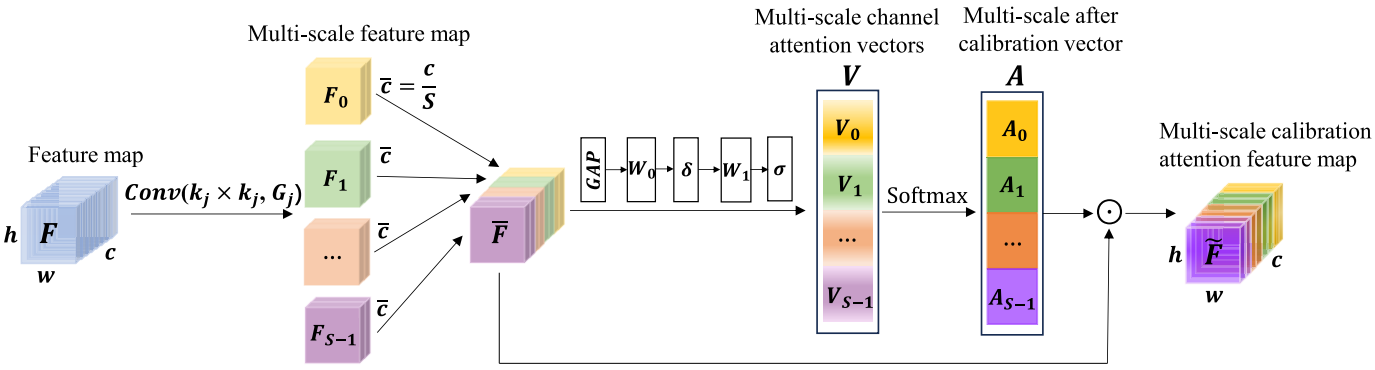


Fig. 5. Structure of MBA module. F denotes the input feature map, F_j denotes multiscale feature map for different branches, $\bar{F}$ is entire preprocessed multiscale feature map. GAP represents global average pooling, δ denotes the linear correction function, w_0 and w_1 denote the fully connected layers, σ denotes the Sigmoid function, V is multiscale channel attention vector, A is calibrated multiscale attention vector.

After that, Softmax is used to independently choose diverse spatial scales by cross-channeling under the guidance of V_j. Then, fusion and splicing are performed to obtain the calibrated multiscale attention vector A that includes the whole spatial location information and cross-channel attention weights. The expression is as follows:

$$A_j = \text{Softmax}(V_j) = \frac{\exp(V_j)}{\sum_{j=0}^{S-1}\exp(V_j)}$$
$$A = A_0 \oplus A_1 \oplus \cdots \oplus A_{S-1}. \tag{4}$$

Then, we multiply the calibrated attention vector A_j of each branch with the F_j of the corresponding scale, and perform the final feature map refinement output. The whole process can be expressed as

$$\tilde{F}_j = F_j \odot A_j$$
$$\tilde{F} = \text{Cat}\left(\left[\tilde{F}_0, \tilde{F}_1, \ldots, \tilde{F}_{S-1}\right]\right) \tag{5}$$

where $\odot$ denotes the elemental multiplication, $\tilde{F}$ denotes the multiscale calibrated attentional feature map containing multi-scale spatial information, as well as cross-channel attentional weights, which enable obtaining more informative interactions between local and global attention.

It is worthwhile noting that $\tilde{F}$ is only the multiscale calibrated attention feature map generated by the MBEN network in one layer of the MBA module. Subsequently, the MBEN module extracts more representative feature maps $f_\theta(\cdot)$ for the SAR image samples by fusing the multiscale calibrated attentional feature maps from different layers to strengthen the differentiation of the feature representations.

C. BC Module

The fundamental idea of the metric-based FSL approach is to achieve category classification by measuring the distance between the support class and the queries. Among them, the metric classification module is an indispensable component of the metric-based FSL learning method. Since there is a scarcity of training samples of SAR images, a single classification criterion is, however, not robust enough for reasoning about target categories in few-shot tasks, and it is difficult to represent the category distribution effectively. If samples can be well categorized simultaneously by different metric modules, then similar samples can be more compactly distributed in the feature space, resulting in more differentiable features. In order to recognize target categories more efficiently and accurately, we design a BC module, which is divided into two main parts: an IC and module a DC module.

Descriptor level is a spatial location in the same channel, which refers to the way of characterizing local features or local information in an image, and captures a small range of visual features by encoding local regions or key points in an image. DC module focuses on the use of image local features (descriptors) for classification, and mainly performs classification operations by analyzing and integrating local descriptors in images.

Specifically, first, in a C-way K-shot task ($D^s = \{(x_i, y_i)\}_{i=1}^{\text{CK}}$, $D^q = \{(q_i, y_i^q)\}_{i=1}^{\text{CQ}}$), D^s (D^q) denotes support set (query set), where $D^s \cap D^q = \phi$. x_i (q_i) and y_i (y_i^q) are the ith support (query) image and its corresponding label. CK (CQ) are the support (query) image numbers; moreover, $x_i \in D^s$ and $q_i \in D^q$ are simultaneously sent to the MBEN module to produce the support feature map $f_\theta(x_i)$ and the query feature map $f_\theta(q_i)$ with representative features. Subsequently, they are fed into the IC module, where the mth support class prototype P_m represents the mean value of $f_\theta(x_i)$ belonging to that class

$$\mathcal{P}_m = \frac{1}{K}\sum_{x_i \in S_m} f_\theta(x_i) \tag{6}$$

where S_m is mth class sample set labeled in D^s. The distance is computed among $f_\theta(q_i)$ and $f_\theta(x_i)$ to represent the prediction score of the q_i predicted to be the mth class as follows:

$$P_{\text{IC}}(y = m|q_i) = \frac{\exp(-d_{\text{eu}}(f_\theta(q_i), \mathcal{P}_m))}{\sum_{m=1}^{C}\exp(-d_{\text{eu}}(f_\theta(q_i), \mathcal{P}_m))} \tag{7}$$

where d_{eu} denotes the Euclidean distance metric.

Subsequently, according to the IC module for each task, the cross-entropy loss function is able to be represented as follows:

$$L_{\text{IC}} = -\frac{1}{\text{CQ}}\sum_{i=1}^{\text{CQ}}\log(P_{\text{IC}}(y = m|q_i)). \tag{8}$$

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Because of the scarcity of samples in each class, such image-level features do not effectively discriminate the elementary distribution of every class. As shown in [48], descriptor-level feature representations are much richer than image-level feature representations and can alleviate the SAR image sample scarcity problem. We, therefore, adopt a DC module that uses deep descriptors for image representation and captures the relationship between the queries and the support classes from both local and global aspects.

Specifically, in task (D^s, D^q), we input the representative feature maps $f_\theta(\cdot)$ are generated by MBEN network to the DC module, a query feature map $f_\theta(q_i)$ and the mth support class feature maps $f_\theta(X_m)$ are given, in which each feature map is able to be regarded as a $c \times h \times w$ 3-D tensor, or as a group of c dimensional depth descriptors.

$$\begin{aligned} f_\theta(q_i) &= \left[q_i^1, \ldots, q_i^n\right] \in \mathbb{R}^{c\times n} \\ f_\theta(X_m) &= \left[f_\theta(x_{m1}), \ldots, f_\theta(x_{mK})\right] \in \mathbb{R}^{c\times nK} \end{aligned} \tag{9}$$

where q_i^l and $f_\theta(x_{ml})$ denote the lth query depth descriptor and the lth support feature map in $f_\theta(X_m)$, respectively, and $n = h \times w$ is the entire depth descriptors. These depth descriptors can be regarded as local representations of the query and support image in the space. For each query image, its distribution in $\mathbb{R}^c$ space is estimated using the n extracted depth descriptors. All depth descriptors of each support class are used together to compute the distribution in $\mathbb{R}^c$ space.

We use an image-to-class metric [48] to capture the local relationship between the query image q_i and the mth support class image X_m. The metric uses depth descriptors as local feature representations of q_i and X_m, and constructs a second-order connected space for both, which can also be viewed as a second-order feature composed by a linear-nonlinear combinatorial calculation. Every element in this space is the dot product among the query sample, as well as the support class sample, which represents the similarity of the two vectors; therefore, the similarity of the elements in this second-order connected space is analyzed to confirm whether the query sample belongs to this support class or not, and this space is further simplified through Topk. Then, the local relationship between q_i and X_m can be expressed as

$$d_{\text{loc}}(q_i, X_m) = \sum_{l=1}^{n} \text{Topk}\left(\frac{f_\theta(q_i)^T \cdot f_\theta(X_m)}{\left\|f_\theta(q_i)^T\right\| \cdot \|f_\theta(X_m)\|}\right) \tag{10}$$

where $\text{Topk}(\cdot)$ indicates the choice of the k maximal elements in every row of the correlation matrix between q_i and X_m. Typically, k is set to 1 in our work, and $\|\cdot\|$ refers to the value of the paradigm.

After that, we use a divergence-distribution metric [34] to capture the global distribution-level relationship between q_i and X_m. Since the distribution of the depth descriptors extracted from q_i or X_m is consistent with the multivariate Gaussian distribution [34] $\mathbb{X} \sim N(\mu_{\mathbb{X}}, \Sigma_{\mathbb{X}})$, then the similarity between them can be computed by combining the mean vector $\mu \in \mathbb{R}^c$, as well as the covariance matrix $\Sigma \in \mathbb{R}^{c\times c}$. The global relationship between q_i and X_m can be defined as follows:

$$\begin{aligned} &d_{\text{glo}}(q_i||X_m) \\ &= \frac{1}{2}\left(c - \text{tr}\left(\Sigma_{f_\theta(X_m)}^{-1}\Sigma_{f_\theta(q_i)}\right) - \ln\left(\frac{\det\Sigma_{f_\theta(X_m)}}{\det\Sigma_{f_\theta(q_i)}}\right)\right. \\ &\qquad \left. -\left(\mu_{f_\theta(X_m)} - \mu_{f_\theta(q_i)}\right)^T\Sigma_{f_\theta(X_m)}^{-1}\left(\mu_{f_\theta(X_m)} - \mu_{f_\theta(q_i)}\right)\right) \end{aligned} \tag{11}$$

where $\text{tr}(\cdot)$ indicates the trace calculation of a matrix, $\ln(\cdot)$ represents the logarithm with e as the base, and det denotes the determinant of a matrix.

Next, we use a fusion layer to fuse the global and local relations between q_i and X_m. The predicted score between q_i and X_m can be represented as below

$$P_{\text{DC}}(y = m|q_i) = \text{FC}\left(\text{BN}\left(\text{Cat}\left(d_{\text{glo}}(q_i||X_m), d_{\text{loc}}(q_i, X_m)\right)\right)\right). \tag{12}$$

As exhibited in Fig. 2, for the three-way one-shot task, the outputs of d_{glo} and d_{loc} are each a 3-D similarity vector. Cat denotes that the two vectors are concatenated together to acquire a 6-D vector. BN denotes that a batch normalization layer is added to poise the scales of the two parts of the similarity. FC denotes a fully connected layer, which is composed of a 1×1 convolutional layer with an expansion value of 3. By using the FC, a weighted 3-D similarity vector is obtained adaptively.

Subsequently, the cross-entropy loss function based on the DC module for every task is able to be formulated as follows:

$$L_{\text{DC}} = -\frac{1}{\text{CQ}}\sum_{i=1}^{\text{CQ}}\log(P_{\text{DC}}(y = m|q_i)). \tag{13}$$

Since the BC module contains two relation scores, i.e., P_{IC} of the IC module and P_{DC} of the DC module, the predicted probability that the query image q_i belongs to the mth class can be finally expressed as

$$\begin{aligned} P(y = m|q_i) &= \lambda_1 * P_{\text{IC}}(y = m|q_i) + \lambda_2 * P_{\text{DC}}(y = m|q_i) \\ \hat{\lambda} &= \frac{\lambda_i}{\sum_{j=1}^{2}\lambda_j}, \quad (i = 1, 2) \end{aligned} \tag{14}$$

where λ_1 and λ_2 are the normalized weights corresponding to P_{IC} and P_{DC}, which are listed in (14). In addition, λ_1 and λ_2 both have an initial value of 1/2, which is used in this article for all experiments.

Finally, the whole model uses a linear combination of these two loss functions to update the gradient.

$$L = L_{\text{IC}} + \alpha * L_{\text{DC}} \tag{15}$$

where α is the trade-off hyperparameter. The model enables to be end-to-end trained through optimizing L using a stochastic gradient descent (SGD) algorithm.

In addition, since the IC module uses image-level feature representation and the DC module uses descriptor-level feature representation, the descriptor-level feature representation is much richer than the image-level feature representation. Accordingly, the ancillary restriction intensity of L_{DC} should be larger, then the weight ratio ought to be larger. The following ablation experiment also verifies our claim.

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TABLE I
PARTITION OF THE FUSAR-SHIP DATASET

Dataset	Ship category	Number
Training set	Cargo	693
	Others	607
	Fishing	785
	Dredger	56
	Tug	54
	Unspecified	138
Testing set	Container	65
	Tanker	162
	BulkCarrier	254

TABLE II
PARTITION OF THE MASTER-SHIP DATASET

Dataset	Ship category	Number
Training set	BMP2	200
	BTR60	200
	BTR70	200
	D7	200
	T62	200
	T72	200
	ZIL131	200
Testing set	2S1	200
	BRDM2	200
	ZSU23_4	200

TABLE III
PARTITION OF THE OPENSAR-SHIP DATASET

Dataset	Ship category	Number
Training set	Cargo	800
	Dredging	374
	Fishing	414
	Tanker	800
	Tug	311
Testing set	Law Enforcement	75
	Passenger	193
	Pilot Vessel	57

III. EXPERIMENTS

A. Experimental Preparation

1) Dataset: For the sake of validating the availability of MBEN-BC method, the experiments related to few-shot SAR images classification using the public SAR ship datasets FUSAR-Ship dataset [49], MSTAR dataset [50] and OPENSAR-Ship dataset [51] are conducted. Following the general setup in FSL, we selected some of the categories for our experiments. All datasets are described in detail in the following three paragraphs.

The FUSAR-Ship dataset is a high-resolution SAR ship dataset containing more than 10 000 SAR image slices, each with a size of 512 × 512 pixels, with 15 ship categories and 98 ship subcategories. This dataset contains images of ships from different viewpoints and environmental conditions, and exhibits relatively large variations in ship target sizes, thus bringing stricter demands on the ability of multiscale feature extraction. In our experiments, all image sizes are cropped to 84 × 84 pixels in order to reduce the redundant background of SAR ship images. Table I gives specific information about every target category for both training and testing data. Typical optical images and the matching SAR images for every category are given in Fig. 6.

To further validate the generalizability of the MBEN-BC method, experiments are conducted on the MSTAR dataset. This dataset is established by the US Air Force Research Laboratory (AFRL). The SAR images have a distance and azimuthal resolution of 0.3 × 0.3 m, and each SAR image size is 128 × 128 pixels. This dataset includes ten categories of military vehicles, i.e., tanks T62, T72; armored personnel carriers BMP2, BRDM2, BTR60, BTR70; air defense units ZSU23/4; rocket launchers 2S1; bulldozers D7; trucks ZIL131; the dataset also has the image data acquired from different angles. Referring to some of the existing popular methods [44], [45], [46], we selected seven types of targets (i.e., T62, BMP2, BTR70, T72, ZIL131, D7, and BTR60) obtained at a 17° pitch angle as the training data, while another three types of vehicles (i.e., ZSU23/4, BRDM2, and 2S1) are used as the test data, and for each type of target, we randomly 200 samples were selected. As in the above method, all image sizes were cropped to 64 × 64 pixels. Table II lists the detailed information of the dataset.

The OPENSAR-Ship dataset is a publicly available high-resolution SAR ship dataset containing more than 15 000 slices of SAR imagery, each of which is 256 × 256 pixels in size, covering 12 major classes of maritime targets and more than 50 subclasses. The dataset contains ship images taken under different weather, sea conditions, and viewpoints, and the shape, size, and azimuth of ship targets vary greatly, making it suitable for multiscale and multiviewpoint feature extraction tasks. In our experiments, in order to remove the redundant background information in the SAR images, all images are cropped to 64 × 64 pixels to focus on the target area. Table III lists the detailed information of each target category in the training and test datasets.

2) Experimental Setup: We use standard data enhancements for all benchmarks in the training phase, including randomly resized cropping, horizontal flipping, and color dithering. The SGD algorithm is adopted, which has a momentum of 0.9 and a weight decay of 5×10^{-4}. Uniform sampling is used for each task in the dataset. We pretrain the model on the training set to obtain initialized weights. Three categories are randomly selected, each with 1 or 5 support images and 15 query images, i.e., three-way one-shot (five-shot). In the testing phase, uniform task numbers are taken from the testing set to evaluate model efficiency. As in the training phase, 15 query images are selected and set to three-way one-shot (five-shot). The average precision of a total of 2000 episodes is recorded, and a 95% confidence interval is used. All the experiments are conducted by adopting the PyTorch and the NVIDIA GeForce RTX3090Ti CPUs.

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TABLE IV
COMPARISON OF CLASSIFICATION OA WITH SUPERIOR METHODS ON THE FUSAR-SHIP DATASET

Method	Source	Backbone	3-way 1-shot	3-way 5-shot
ProtoNet † [32]	NeurIPS'17	Conv4	65.93±0.51	75.78±0.30
RelationNet [33]	CVPR'18	Conv4	65.27±0.68	74.76±0.30
DN4† [49]	CVPR'19	Conv4	67.07±0.43	75.70±0.28
MATANet [41]	ICPR'22	Conv4	67.33±0.60	77.02±0.45
Meta DeepBDC [37]	CVPR'22	Conv4	68.40±0.50	76.32±0.29
LATA-BP-BDC [40]	TGRS'23	Conv4	68.75±0.45	78.35±0.28
Ours	Ours	Conv4	**69.71±0.45**	**79.88±0.28**

TABLE V
COMPARISON OF CLASSIFICATION OA WITH SUPERIOR METHODS ON THE MSTAR DATASET

Method	Source	Backbone	3-way 1-shot	3-way 5-shot
ProtoNet † [32]	NeurIPS'17	Conv4	69.71±0.55	83.57±0.28
RelationNet [33]	CVPR'18	Conv4	69.81±0.66	81.55±0.28
DN4† [49]	CVPR'19	Conv4	71.51±0.51	83.71±0.30
MATANet [41]	ICPR'22	Conv4	75.13±0.47	88.68±0.53
Meta DeepBDC [37]	CVPR'22	Conv4	73.72±0.55	88.49±0.24
LATA-BP-BDC [40]	TGRS'23	Conv4	75.18±0.53	89.66±0.25
Ours	Ours	Conv4	**78.04±0.51**	**91.21±0.20**

TABLE VI
COMPARISON OF CLASSIFICATION OA WITH SUPERIOR METHODS ON THE OPENSAR-SHIP DATASET

Method	Source	Backbone	3-way 1-shot	3-way 5-shot
ProtoNet † [32]	NeurIPS'17	Conv4	41.78% +- 0.42%	49.60% +- 0.34%
RelationNet [33]	CVPR'18	Conv4	39.76%+- 0.41%	48.81%+- 0.36%
DN4† [49]	CVPR'19	Conv4	44.93%+-0.48%	49.52%+- 0.35%
MATANet [41]	ICPR'22	Conv4	44.31%+- 0.67%	50.95%+- 0.49%
Meta DeepBDC [37]	CVPR'22	Conv4	42.70%+- 0.47%	50.55%+- 0.35%
LATA-BP-BDC [40]	TGRS'23	Conv4	45.18% +- 0.47%	51.07%+- 0.34%
Ours	Ours	Conv4	**45.65%+-0.49%**	**52.12%+-0.36%**

B. Comparison With Superior Methods

Since the MBEN-BC method is a metric-based few-shot method, the FUSAR-Ship, MSTAR and OPENSAR-Ship datasets are used to compare the MBEN-BC with other currently popular and advanced metric-based few-shot methods, including: prototypical networks (ProtoNet) [31], RelationNet [32], DN4 [48], MATANet [40], Meta DeepBDC [36], and lightweight adaptive task attention-Bi-prototype-Brownian distance covariance (LATA-BP-BDC) [39].

Tables IV–VI report the three-way classification overall accuracy (OA) [53] for the FUSAR-Ship dataset, the MSTAR dataset, and the OPENSAR-Ship dataset, respectively, for different methods using the backbone network Conv4. Methods with "†" indicate that our training pattern and backbone were used to reproduce the model. From Table IV, it can be clearly seen that the MBEN-BC method gains the best result for both one-shot and five-shot settings, with an increase of 0.96% (one-shot) and 1.53% (five-shot) compared with LATA-BP-BDC, respectively. Table V also shows similar results on the MSTAR dataset. Compared with LATA-BP-BDC, the MBEN-BC method improves by 2.86% (one-shot) and 1.55% (five-shot), respectively. Table VI also shows similar results on the OPENSAR-Ship dataset. Compared to LATA-BP-BDC, the MBEN-BC method improved by 0.47% (1 time) and 1.05% (5 times), respectively. We, therefore, conclude that MBEN-BC method achieves the best performance for both the complex and unevenly distributed FUSAR-Ship dataset and OPENSAR-Ship dataset and the uniformly distributed MSTAR dataset, which demonstrates the powerful classification capability of MBEN-BC method.

C. Ablation Study

1) MBEN Module: In this portion, we perform ablation experiments on the FUSAR-Ship, the MSTAR dataset, and the OPENSAR-Ship dataset to certify the availability of the MBEN module. First, without changing other parameter settings, the Conv4 and MBEN are used as feature embedding networks, respectively, to confirm the advantage of multiscale feature extraction in few-shot SAR image classification. Since the MBEN mainly consists of several MBA modules, 0–3 MBA modules are used for the ablation experiments, where the number of MBA modules is 0, indicating the Conv4. Tables VII–IX show the ablation results for different numbers of MBA modules on the FUSAR-Ship, the MSTAR dataset, and the OPENSAR-Ship dataset, respectively. It can be clearly seen that as the number of MBA modules adds, good performance is achieved both in the three-way one-shot

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TABLE VII
COMPARISON OF DIFFERENT MBA MODULE NUMBERS ON THE FUSAR-SHIP DATASET

Numbers of MBA	3-way 1-shot	3-way 5-shot
0	65.93±0.51	75.78±0.30
1	66.69±0.48	77.88±0.29
2	67.57±0.52	78.07±0.27
3	**68.20±0.49**	**78.98±0.28**

TABLE VIII
COMPARISON OF DIFFERENT MBA MODULE NUMBERS ON THE MSTAR DATASET

Numbers of MBA	3-way 1-shot	3-way 5-shot
0	69.71±0.55	83.57±0.28
1	72.37±0.59	86.69±0.24
2	**77.23±0.51**	89.64±0.22
3	76.79±0.51	**89.82±0.22**

TABLE IX
COMPARISON OF DIFFERENT MBA MODULE NUMBERS ON THE OPENSAR-SHIP DATASET

Numbers of MBA	3-way 1-shot	3-way 5-shot
0	42.68±0.36	49.74±0.36
1	45.47±0.47	51.42± 0.37
2	45.45± 0.47	51.91± 0.35
3	**45.50± 0.48**	**52.02±0.36**

(five-shot) setting. For the FUSAR-Ship dataset, the MBEN module improved performance by 2.27% (one-shot), as well as 3.2% (five-shot) compared with the Conv4. For the MSTAR dataset, the efficiency improved by 7.08% (one-shot) as well as 6.25% (five-shot). For the OPENSAR-Ship dataset, the efficiency improved by 2.82% (one-shot), as well as 2.28% (five-shot). In the one-shot setting for the MSTAR dataset, the precision of MBA module number 2 is a little higher than that of module number 3. The reason for analyzing this may be that the MSTAR dataset is uniformly distributed and the location of each image is in the middle, which is less complex compared with the FUSAR-Ship dataset. The network with MBA module number 2 is, therefore, sufficient to obtain discriminative features. In general, the MBEN module is, however, able to achieve good performance on both the unevenly distributed and relatively complex FUSAR-Ship dataset and the uniform MSTAR dataset.

2) BC Module: Tables X–XII show the ablation results of the BC module on the FUSAR-Ship dataset, the MSTAR dataset, and the OPENSAR-Ship dataset, respectively. For the sake of evaluating the availability of the BC module, we designed two groups of experiments, respectively; the first group of experiments for validating the BC against the baseline network "ProtoNet," with experimental data in the first four rows of the table, and another set of experiments with the addition of the "MBEN" module, and the experimental data are in the last four rows of the table. In the first set of experiments, "ProtoNet-L" denotes that only local relations are captured using the image-to-class metric in the DC module, and "ProtoNet-G" denotes that only global relations

TABLE X
ABLATION STUDY OF BC MODULE ON THE FUSAR-SHIP DATASET

Method	3-way 1-shot	3-way 5-shot
ProtoNet	65.93±0.51	75.78±0.30
ProtoNet-L	66.29±0.49	**77.71±0.27**
ProtoNet-G	66.23±0.48	77.13±0.27
ProtoNet-BC	**67.07±0.47**	77.29±0.28
MBEN	68.20±0.49	78.98±0.28
MBEN-L	68.72±0.45	79.12±0.27
MBEN-G	68.30±0.46	79.58±0.29
Ours	**69.71±0.45**	**79.88±0.28**

TABLE XI
ABLATION STUDY OF BC MODULE ON THE MSTAR DATASET

Method	3-way 1-shot	3-way 5-shot
ProtoNet	69.71±0.55	83.57±0.28
ProtoNet-L	70.75 ± 0.54	86.66±0.26
ProtoNet-G	70.24±0.51	86.01±0.30
ProtoNet-BC	**72.56±0.52**	**87.14±0.26**
MBEN	76.79±0.51	89.82±0.22
MBEN-L	77.45±0.49	**91.65±0.21**
MBEN-G	77.79±0.55	90.15±0.29
Ours	**78.04±0.51**	91.21±0.20

TABLE XII
ABLATION STUDY OF BC MODULE ON THE OPENSAR-SHIP DATASET

Method	3-way 1-shot	3-way 5-shot
ProtoNet	41.78±0.42	48.60±0.34
ProtoNet-L	41.93±0.48	49.52±0.35
ProtoNet-G	42.50±0.41	50.07 ± 0.35
ProtoNet-BC	**42.58 ± 0.48**	**50.25±0.36**
MBEN	42.68±0.36	49.74±0.36
MBEN-L	44.35±0.47	51.83±0.38
MBEN-G	44.99±0.48	50.60±0.35
Ours	**45.65±0.49**	**52.12±0.36**

are captured using the scatter distribution metric in the DC module. "ProtoNet-BC" denotes that the entire BC module is added to the baseline. "MBEN-L" denotes that only local relationships are captured using the image-to-class metric in the DC module, and "MBEN-G" denotes that only global relationships are captured using the scatter distribution metric in the DC module. "Ours" denotes that the entire BC module is added to the MBEN.

In Table X, for the FUSAR-Ship dataset, the experimental results show that the ProtoNet network outperforms the single classifier module by 0.36%, 0.30%, and 1.14% (1.93%, 1.35%, and 1.51%) for the three-way, one-issue (five-issue) setups for the capturing of local relationships only, of capturing global relationships, and the overall BC module, respectively. The latter four experiments showed that capturing only local relations, only global relations, and the overall BC module improved the MBEN network by 0.52%, 0.1.%, and 1.51% (0.14%, 0.60%, and 0.90%), respectively, in the three-way, one-shot (five-shot) setting. In Table XI, for the MSTAR dataset, the "ProtoNet-L," "ProtoNet-G," and "ProtoNet-BC" networks enhance the accuracy over the ProtoNet network in

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TABLE XIII
ABLATION STUDY OF DIFFERENT α VALUES ON THE FUSAR-SHIP DATASET

Value of α	0	0.01	0.05	0.1	0.5	1	5	10
3-way 1-shot	68.20	68.78	68.58	69.21	69.31	**69.71**	69.47	69.10
3-way 5-shot	78.98	79.15	79.34	79.71	79.67	79.85	**79.88**	79.54

TABLE XIV
ABLATION STUDY OF DIFFERENT α VALUES ON THE MSTAR DATASET

Value of α	0	0.01	0.05	0.1	0.5	1	5	10
3-way 1-shot	76.79	77.11	77.09	77.04	77.84	**78.04**	77.79	76.84
3-way 5-shot	89.82	90.31	90.28	90.33	90.69	90.95	**91.21**	91.13

the three-way one-shot (five-shot) setting by 1.04%, 0.53%%, and 2.85% (3.09%, 2.44%, and 3.57%). The "MBEN-L," "MBEN-G," and "MBEN-BC" networks have, moreover, improved performance over the MBEN network in the one-shot (five-shot) setting by 0.66%, 1.00%, and 1.25% (1.83%, 0.33%, and 1.39%). In Table XII, for the OPENSAR-Ship dataset, the "ProtoNet-L," "ProtoNet-G," and "ProtoNet-BC" networks enhance the accuracy over the ProtoNet network in the three-way one-shot (five-shot) setting by 0.15%, 0.72%%, and 0.80% (0.92%, 1.47%, and 1.65%). The "MBEN-L," "MBEN-G," and "MBEN-BC" networks have, moreover, improved performance over the MBEN network in the one-shot (five-shot) setting by 1.67%, 2.31%, and 2.97% (2.09%, 0.86%, and 2.38%).

Capturing local relationships using image-to-class metrics, therefore, effectively increases the sensitivity of local features to classification. Capturing global relationships using the divergence-distribution metric allows analyzing and evaluating the distribution of different categories in the entire feature space increasing the generalization ability of the model. The BC module constructed by fusing the two can induce robust target category inference in a small number of tasks with two categorization criteria, thus effectively representing the category distribution.

3) Choice of Hyper-Parameters: The value of α represents the trade-off hyperparameter of the loss function, which is used to control the relative importance of the primary and auxiliary loss functions, and by adjusting the value of α, the model can find the best balance between the primary and auxiliary tasks. The effect of its value on the classification effect will be analyzed in several experiments below. We set $\alpha = 0$, 0.01, 0.05, 0.1, 0.5, 1, 5, and 10, where $\alpha = 0$ means that the BC module is not used, in which case the DC module does not play a role at all, and the model focuses only on the loss of the IC module. While $0 < \alpha < 1$, the IC module is still dominant, and the strength of auxiliary constraints of the DC module increases with α, which can increase the model's generalization ability but does not significantly change the optimization direction of the main task. When $\alpha = 1$ indicates that the loss functions of IC and DC modules are equally important, the model is able to consider the two classification modules at the same time, which improves the performance or accelerates the convergence speed. When $\alpha > 1$, the constraint strength of DC module is larger, and an appropriate increase of α value helps to better extract complex features in some complex tasks.

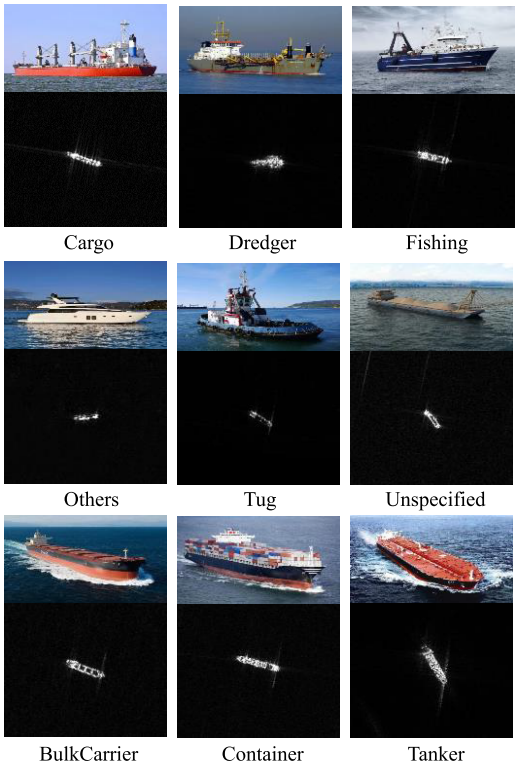


Fig. 6. SAR ships and matching optical ships in the FUSAR-Ship dataset.

The classification result of the MBEN-BC network on the FUSAR-Ship, the MSTAR dataset, and the OPENSAR-Ship dataset are exhibited in Tables XIII–XV, respectively. The variation curves of classification accuracy on three datasets with different α values respectively are, moreover, provided in Fig. 7. It can be seen that the MBEN-BC network achieves the highest precision with $\alpha = 1$ ($\alpha = 5$) for the three-way one-shot (five-shot) setting; therefore, a reasonable setting of α is needed, and in this article, $\alpha = 1$ ($\alpha = 5$) under the three-way one-shot (five-shot) setting is suggested. We explore the adjustment of alpha to the loss function of the classifier; the effect of alpha is the same when using different datasets, and the experimental results also illustrate that the optimal value of alpha is the same.

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TABLE XV
ABLATION STUDY OF DIFFERENT α VALUES ON THE OPENSAR-SHIP DATASET

Value of α	0	0.01	0.05	0.1	0.5	1	5	10
3-way 1-shot	44.69	44.69	44.55	44.59	45.25	**45.65**	45.27	45.24
3-way 5-shot	51.31	51.57	51.38	51.29	51.90	52.11	**52.12**	51.03

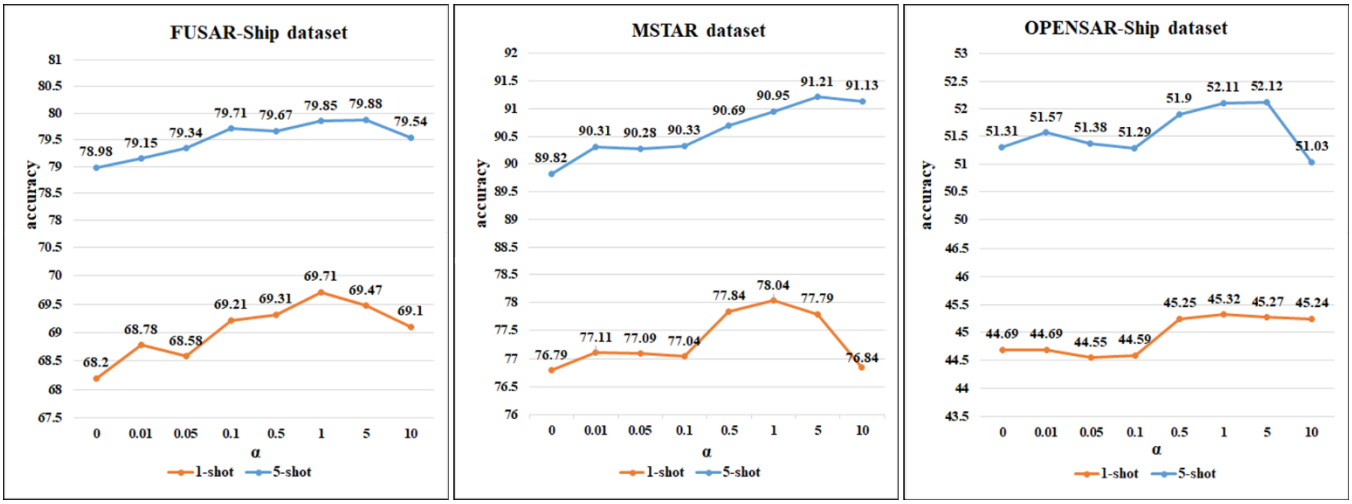


Fig. 7. Variation of classification accuracy with different α values on the three datasets. (a) FUSAR-Ship dataset. (b) MSTAR dataset. (c) OPENSAR-Ship dataset.

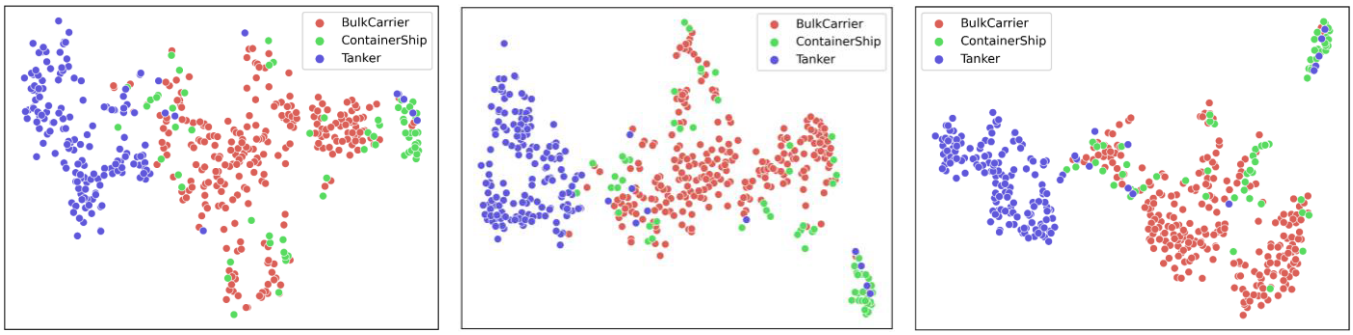


Fig. 8. Feature space of different networks on the SAR ship target (FUSAR-Ship dataset). (a) bBseline network. (b) MBEN network. (c) MBEN-BC network.

D. Visualization

For the sake of demonstrating more intuitively the effectiveness of the MBEN-BC network in improving the performance of image target recognition, we qualitatively analyze from the point of view of sample feature distribution and adopt t-distributed stochastic neighbor embedding (T-SNE) [52] to visualize the feature embeddings of the original SAR image, the ProtoNet network feature embeddings, and the MBEN-BC network, respectively. Figs. 8 and 9 show the embedding distributions of the three types in the SAR ship target FUSAR-Ship (OPENSAR-Ship results are similar and not shown to save spaces) and vehicle target MSTAR datasets, respectively. From Figs. 8(a) and 9(a), it is clear that the distribution of raw SAR images in each class is extremely scattered, and there is even a serious overlap between classes. From Fig. 8(b) and (c) and Fig. 9(b) and (c), it is obvious that the visualization results of the ProtoNet network and the MBEN-BC network have better interclass differentiation than those represented in Figs. 8(a) and 9(a). In addition, the MBEN-BC network can obtain a more compact intraclass space and a more separable interclass space than the ProtoNet network. The MBEN-BC network can, therefore, improve the efficiency and robustness of the model by enhancing the distinguishability of the feature space.

Here, we have a point to add that the similarity distribution in Figs. 8 and 9 shows a better classification. We notice that since the OA is, however, limited in the previous Tables IV–VI, in combination with [53] we can explain the difference well. The potential of transfer learning-based approaches (e.g., SPT-FSC proposed in [53]) to enhance model performance in challenging and complex visually disparate scenarios is more capable of achieving accurate and robust target recognition, and accordingly, more significant in improving classification accuracy. On the contrary, meta-learning-based approaches (e.g., MBEN-BC in this article) have a comparable ability to learn discriminative features from a small number of samples, i.e., a better judgment of the spatial distribution of the samples.

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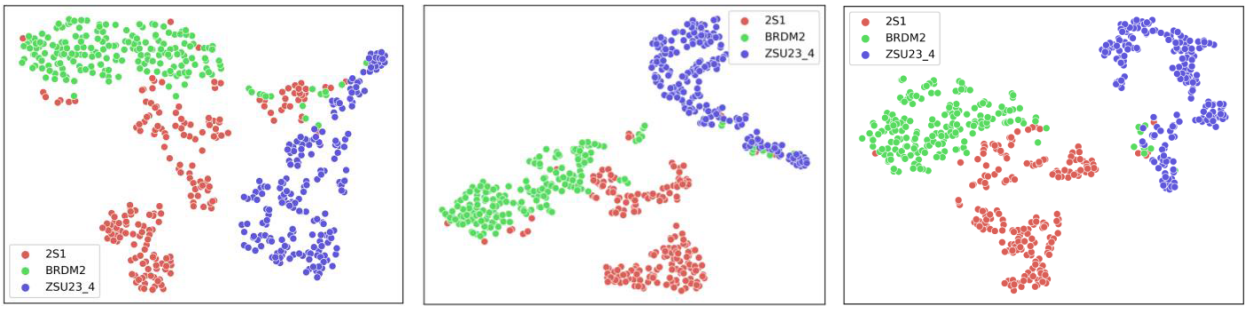


Fig. 9. Feature space of different networks on the vehicle target (MSTAR dataset). (a) Baseline network. (b) MBEN network. (c) MBEN-BC network.

IV. Conclusion

MBEN-BC network was proposed to realize FSC for SAR image targets. In terms of feature extraction, the MBEN module captured multiscale spatial information by extracting image features at different scales in a multibranching manner. Meanwhile, the cross-channel attention mechanism enriched the amount of information in the feature space and effectively extracted discriminative features by establishing information interaction between local and global attention. In the classification stage, the BC module constructed classification criteria from the image level and descriptor level, respectively, which promoted the similar samples to be distributed more compactly in the feature space and improved the network's ability to represent scarce labeled samples. In this article, experiments were conducted on the FUSAR-Ship dataset, the MSTAR dataset, and the OPENSAR-Ship dataset; comparative and ablation experiments were conducted to validate the superior performance of the MBEN-BC network in the task of classifying SAR-image ships, as well as to demonstrate its superior generalization ability on a wider range of SAR datasets.

Ship targets are usually small in SAR images, so the discriminative information that can be extracted is very limited. In addition, the SAR dataset available for training is sparsely sampled. This makes it possible that the use of deeper backbone networks may not be significantly effective in classifying small samples of ships in SAR. In addition, future research could further explore the effects of sample imbalance on the small sample SAR ship classification task, which can adversely affect the generalization ability of the model. Although this issue is not discussed in detail in this article, it is worth noting that there are different approaches to ship classification, such as solving the problem based on data augmentation methods from the perspective of physical attributes and DL and from the perspective of optical knowledge migration for rich target recognition, which may provide a more comprehensive solution for SAR image classification. This article focuses on meta-learning-based approaches to solve problems in feature extraction and metric classification, and future research could further optimize the feature extraction and classification performance in SAR ship classification tasks by better fusing the advantages of different approaches.

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